EXPERIMENT 10
Performance Optimization in Zoho Creator

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Aim
To demonstrate performance optimization techniques in a Zoho Creator application, particularly efficient record retrieval and workflow processing.

Introduction
Performance optimization in low-code applications means designing forms, workflows, and scripts so that they use data and processing only as far as needed. Although a low-code platform handles much of the underlying work, the way in which a workflow is written still influences how efficiently it runs.

Retrieving more records than required, or repeatedly processing records that are not relevant to the task, causes unnecessary data retrieval and processing. In a small application this may not be noticeable, but as the number of records and users grows it can reduce application efficiency. Efficient workflows in Zoho Creator therefore retrieve only the data they need and avoid unnecessary processing, which supports scalable and maintainable applications.

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Application Used
The practical was performed in Zoho Creator 6 using the **Loan Application System**. Deluge scripting was used to write the logic of a form workflow that sends an email notification. The Zoho Creator account was on an expired trial, so some advanced features required an upgrade; no upgrade-only feature is claimed as enabled or configured in this record.

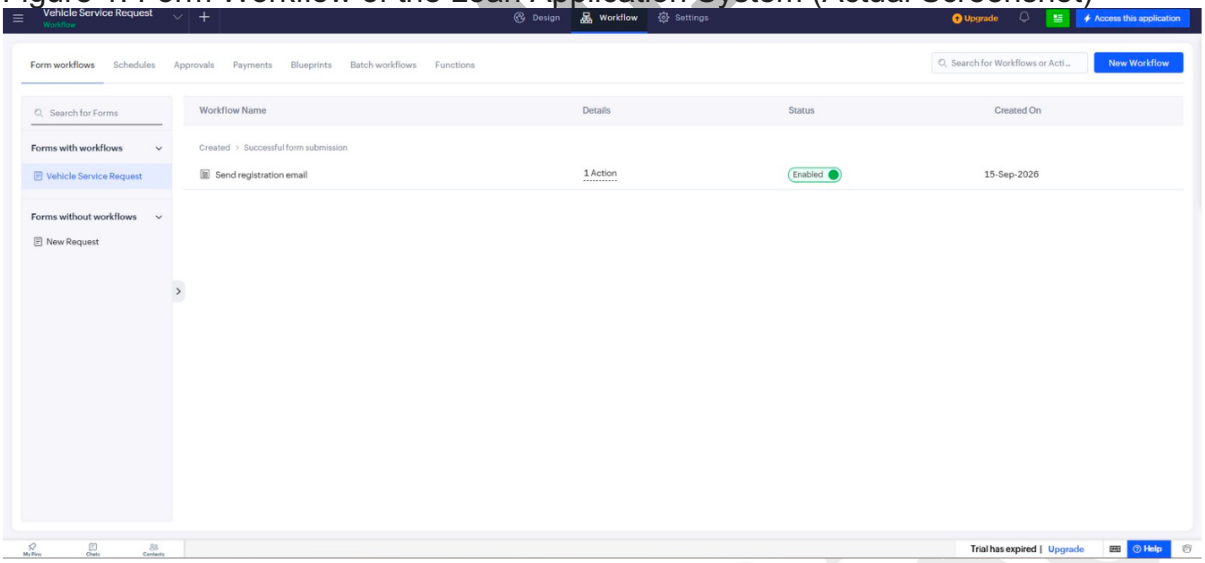
Performance Optimization Concept
The key optimization demonstrated is criteria-based record fetching. Instead of retrieving every record of a form, a criterion is given so that only the required record is returned. In the **Loan Application** workflow, this is written as:

- The workflow identifies the current record using input.ID, which holds the ID of the record being processed.
- Only the relevant Loan Application record, the one whose ID matches input.ID, is retrieved.
- This avoids unnecessarily fetching and processing unrelated records.
- This improves workflow efficiency and reduces unnecessary data processing.

Procedure

1. Open the **Loan Application System** application in Zoho Creator in Edit mode.
2. Navigate to the Workflow section and open Form workflows. The form **Loan Application** is listed under Forms with workflows.
3. Observe the form workflow **"Send Loan Registration Email"**. It is triggered on **Created > Successful form submission**, has 1 action, and has the status **Enabled** (Figure 1).

Figure 1: Form Workflow of the Loan Application System (Actual Screenshot)



4. When a Loan Application is submitted, the workflow identifies the current record using input.ID.
5. The required record is fetched using the criteria ID == input.ID, so that only this record is retrieved.

6. The workflow checks whether the email field of the record is blank using .isBlank().

7. If the email is available, the notification email is sent to the customer. If the email is missing, an informational message is displayed.

Deluge Code

The Deluge code of the optimized workflow is as follows:

Code snippet

```
loanApplication = Loan_Application1[ID == input.ID];
if (loanApplication.email.isBlank())
{
   info "Email address is missing for this request";
}
else
{
   sendmail
   [
      from: zoho.adminuserid
      to: loanApplication.email
      subject: "Your Loan Application is Received"
      message: "<p>Dear Applicant,</p><p>Your loan application has been received and accepted for processing. We will contact you shortly regarding the next steps.</p><p>Thank you for choosing our services.</p>"
```

The first statement fetches the record by its ID. If the email field of that record is blank, an informational message is displayed using info. Otherwise, an email with the subject "Your Loan Application is Received" is sent to the applicant's email address.

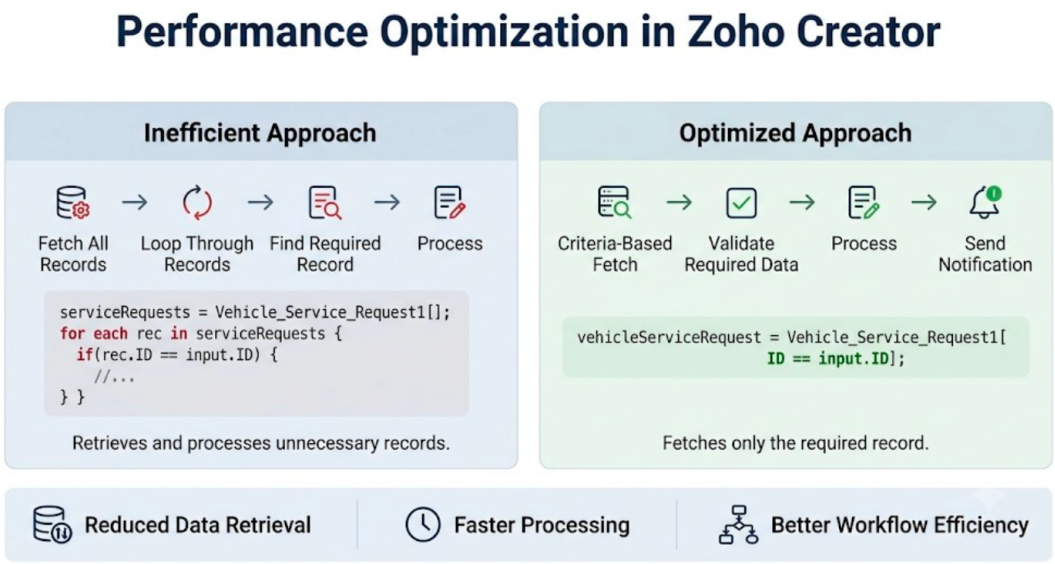
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Performance Analysis

The optimized approach is compared with a general inefficient approach below:

Approach	Steps
Before (inefficient approach)	Fetch all records → Iterate through records → Find required record → Process
Optimized (used in workflow)	Fetch required record using ID → Process directly

Figure 2: Inefficient and Optimized Approaches (Conceptual Diagram)]



Illustrative Comparison Code (Not used in live application)

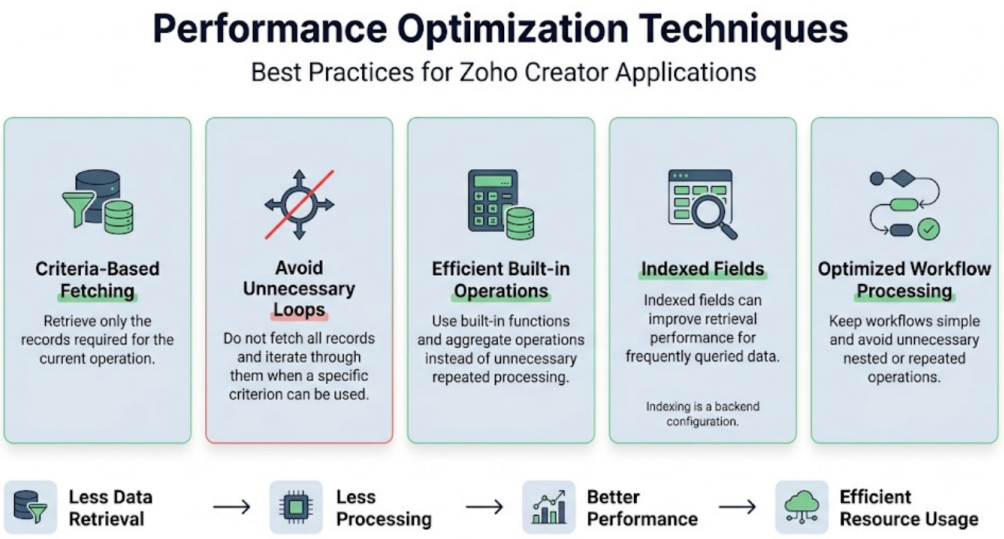
```
// Inefficient approach for illustration only
loanApplications = Loan_Application1[];
for each rec in loanApplications
{
   if (rec.ID == input.ID)
   {
      // Process record...
   }
}
```

The inefficient approach retrieves and processes records that are not needed, whereas the optimized approach reduces unnecessary record retrieval and processing. Figure 2 illustrates this comparison. No benchmark was performed in this practical, so no measured improvement in execution time is claimed. The comparison is based on the amount of data retrieved and processed.

Optimization Techniques

General practices that contribute to performance in Zoho Creator applications are given below:

Figure 3: Performance Optimization Techniques (Conceptual Diagram)



Technique	Explanation
Criteria-based record fetching	Retrieve only the records required for the current operation. (Used in this practical)
Avoiding unnecessary record iteration	Do not fetch all records and loop through them when a specific criterion can be used.
Efficient built-in functions and aggregate operations	Use built-in functions and aggregate operations instead of unnecessary repeated processing.
Indexed fields	Indexed fields can improve retrieval performance for frequently queried data. Indexing may require backend/platform configuration, and it was not enabled or configured in this application.
Efficient workflows	Keep workflows simple and avoid unnecessary nested or repeated operations.

Optimized Workflow

1. **Identify current record:** input.ID gives the ID of the submitted record.
2. **Criteria-based record fetch:** Loan_Application1[ID == input.ID] retrieves only the relevant record.
3. **Validate email:** email.isBlank() checks whether the email field is empty.
4. **Prepare notification and send email:** If the email is available, the notification message is prepared and sent using sendmail; otherwise, an informational message is displayed.

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Conclusion

Performance optimization is important in low-code applications because efficiency depends not only on the platform but also on how workflows and scripts are written. In this practical, the **Loan Application** workflow used criteria-based record retrieval with the expression Loan_Application1[ID == input.ID], so that only the required record was retrieved and processed.

Targeted record retrieval improves efficiency by avoiding the retrieval of unrelated records and iteration through them, and efficient workflow design reduces unnecessary processing. Practices such as criteria-based fetching, avoiding unnecessary loops, using built-in operations, and keeping workflows simple help applications handle growing amounts of data. These practices contribute to scalable and maintainable Zoho Creator applications. No benchmark was performed, so the benefits are described in terms of reduced data retrieval and processing and not as measured execution times.